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Weighted McNemar's test for the comparison of two screening tests in the presence of verification bias

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Abstract

Statistical methods have been well-developed for comparing two binary screening tests in the presence of verification bias. However, the complexity of existing methods and the computational difficulty in implementing them have restricted their use. A simple and easily implemented statistical method is therefore needed. In this paper, we propose a weighted McNemar's test statistic for comparing two sensitivities(specificities). The proposed test statistics are intuitive and simple to compute, only involving some minor modification of a McNemar's test statistic using the estimated verification probabilities for discordant pairs. Simulations demonstrate that the proposed weighted McNemar's test statistics preserve type I error as well as or better than the existing statistics. Furthermore, unlike the existing methods, the proposed weighted McNemar's test statistics can still be applied even when none of the accordant pairs are verified.

KEYWORDS

McNemar's test, sensitivity, specificity, verification bias, weighted McNemar's test

1 | INTRODUCTION

The most accurate clinical methods to diagnose a disease are often not practical for routine screening purposes in the general population. Therefore, screening tests that are less invasive and/or less expensive than the gold standard methods are commonly employed in disease surveillance. If the result of a screening test is binary (positive or negative), then the accuracy of the test can be represented by its sensitivity and specificity.[1] Sensitivity measures how good a test is in detecting a diseased subject, whereas specificity measures how good it is in detecting a nondiseased subject. In the presence of two competing screening tests, it is of interest to compare their the ability to correctly detect a disease as measured by their sensitivities and specificities.

McNemar's test[2] is often used to compare two binary screening tests when both screening tests and the gold standard are available for every study subject in a random sample. In practice, however, often the gold standard is so invasive or costly that only a subset of study subjects receive disease verification and the verification probability differs depending on the results of the two screening tests. For example, one of the aims in an epidemiological study of Alzheimer's disease[3] is to compare the accuracy of a new screening test with that of a standard screening test. The new test[3] is based on information from a cognitive test given to the subject and from a relative test given to someone who knows the subject. The standard test uses results from the cognitive test only.[4] Because of time and cost restrictions, not all screened subjects can be assessed clinically. The probability of selecting a subject for a clinical assessment was based on the results of the screening tests of the subject and on whether the subject's age was less than 75 years. In such a partial disease verification situation, the statistical methods applied directly to the verified subjects may lead to biased results, known as verification bias.[5,6]

Several methods[5-7] have been proposed for correcting for verification bias in estimating the accuracy of a single screening test. Methods also exist for comparing two binary screening tests in the presence of verification bias. Zhou[8] developed a test statistic based on the difference of two bias-corrected estimators of sensitivity(specificity). Application of Zhou's test requires that the verification probabilities for all four combinations of the two binary test results be greater than zero. Alonzo[9] proposed a test statistic based on the ratio of two bias-corrected estimators of sensitivity(specificity). Unlike Zhou's test statistic, Alonzo's test statistic can still be applied when none of subjects who screen negative on both screening tests receive the gold standard. However, the complexity of these two methods and the computational difficulty in implementing them has restricted their use. A simple and easily implemented statistical method is therefore needed. In this paper, we propose a weighted McNemar's test statistic, which only involves some minor modification of a McNemar's test statistic using the estimated verification probabilities for discordant pairs. The proposed method is advantageous over existing methods in that it accommodates for more general verification schemes as long as the verification probabilities for discordant pairs are positive, and more importantly, it is more intuitive and much simpler to compute.

We organize the paper as follows. In Section 2, we give a brief review on the two existing methods for comparing two sensitivities and specificities. We derive the proposed weighted McNemar's test statistic in Section 3. In Section 4, we show that the proposed weighted McNemar's test statistic is more efficient than McNemar's test statistic, which can be applied when the verification probabilities for discordant pairs are equal. The extension of the proposed weighted McNemar's test statistic to adjust for covatiates is presented in Section 5. Simulation studies that compare the empirical sizes and powers of the proposed statistics and existing statistics are summarized in Section 6. Section 7 applies the proposed method to data from the study of Hall et al.[3] comparing the accuracy of two screening tests for detecting Alzheimer's disease. We end with some recommendations and a discussion.

2 | REVIEW OF EXISTING METHODS

This section introduces notations and review two existing methods for comparing two sensitivities and specificities in the presence of verification bias.

Suppose that two binary screening tests are applied independently to a random sample of n subjects. Let D be the true disease status, where $D = 1$ for a diseased subject and $D = 0$ otherwise, and let T_1 and T_2 be the results from screening test 1 and screening test 2, respectively, where $T_k = 1$ for a positive test and $T_k = 0$ for a negative test, $k = 1, 2$. The joint distribution of (T_1, T_2, D) is denoted by

$$p_{uvw} = \Pr(T_1 = u, T_2 = v, D = w), \quad \text{for} \quad u, v, w = 0, 1.$$

Let R be disease verification indicator variable where $R = 1$ for a verified subject and $R = 0$ for a nonverified subject. We assume that the probability of selecting a subject for disease verification depends on only the screening test results of the subject rather than their true disease status, that is, $\Pr(R = 1|T_1, T_2, D)$=$\Pr(R = 1|T_1, T_2)$, and let τ_{uv} denote the true verification probability for a subject with $(T_1 = u, T_2 = v), u, v = 0, 1$.

Assume that binary results from two screening tests are collected from a random sample of n subjects. Let $(R_i, T_{1i}, T_{2i}, D_i)$ be the value of (R, T_1, T_2, D) of the ith subject, respectively, $i = 1, \dots, n$, where T_{1i} and T_{2i} are observed for all subjects, but D_i is observed only if $R_i = 1$. Table 1 summarizes the observed frequency data where n_{uv} and m_{uv} are the numbers of subjects and verified subjects with $(T_1 = u, T_2 = v)$, and r_{uv} and s_{uv} are the numbers of verified subjects with $(T_1 = u, T_2 = v, D = 0)$ and $(T_1 = u, T_2 = v, D = 1)$, respectively, $u, v = 0, 1$.

The aim of inference is to assess the difference in accuracy between two screening tests. Let Se_k and Sp_k be the sensitivity and specificity of the kth screening test respectively, $k = 1, 2$, that is,

$$Se_1 = (p_{111} + p_{101})/\lambda, \qquad Se_2 = (p_{111} + p_{011})/\lambda,$$
$$Sp_1 = (p_{000} + p_{010})/(1 - \lambda), \qquad Sp_2 = (p_{000} + p_{100})/(1 - \lambda),$$

where $\lambda = \Pr(D = 1)$. To assess the differences in sensitivity and specificity, the null hypotheses of interest are $H_0 : Se_1 = Se_2$ and $H_0 : Sp_1 = Sp_2$. Since the null hypothesis $H_0 : Se_1 = Se_2$ is equivalent to $H_0 : p_{101} = p_{011}$, and the null hypothesis of interest is $H_0 : Sp_1 = Sp_2$ is equivalent to $H_0 : p_{100} = p_{010}$, in Section 3, we will derive weighted MeNemar's test statistics to test $H_0 : p_{101} = p_{011}$ and $H_0 : p_{100} = p_{010}$.

TABLE 1 Cross-classification of test results by disease status and verification status for a sample of n patients who undergo two binary screening tests. $D = 1$ for a diseased subject and $D = 0$ otherwise. $T_k = 1$ for if the result of kth screening test is positive and $T_k = 0$ otherwise, $k = 1, 2$. $R = 1$ for a verified subject and $R = 0$ otherwise

	$T_1 = 1$		$T_1 = 0$	
	$T_2 = 1$	$T_2 = 0$	$T_2 = 1$	$T_2 = 0$
$R = 1$				
$D = 1$	s_{11}	s_{10}	s_{01}	s_{00}
$D = 0$	r_{11}	r_{10}	r_{01}	r_{00}
Total	m_{11}	m_{10}	m_{01}	m_{00}
$R = 0$	l_{11}	l_{10}	l_{01}	l_{00}
Total	n_{11}	n_{10}	n_{01}	n_{00}

2.1 | Zhou's test statistics

Zhou[8] proposed test statistics for comparing two sensitivities and specificities in the presence of verification bias. They are constructed based on the difference of the maximum likelihood estimators of two sensitivities and the difference of the maximum likelihood estimators of two specificities. Let $\widehat{Se}_k$ and $\widehat{Sp}_k$ be their respective maximum likelihood estimators of Se_k and $Sp_k, k = 1, 2$. Then $\widehat{Se}_k$ and $\widehat{Sp}_k$ are given by

$$\widehat{Se}_1 = \sum_{v=0}^{1} s_{1v}\hat{\tau}_{1v}^{-1} \Big/ \sum_{u,v=0}^{1} s_{uv}\hat{\tau}_{uv}^{-1}, \qquad \widehat{Se}_2 = \sum_{u=0}^{1} s_{u1}\hat{\tau}_{u1}^{-1} \Big/ \sum_{u,v=0}^{1} s_{uv}\hat{\tau}_{uv}^{-1},$$
$$\widehat{Sp}_1 = \sum_{v=0}^{1} r_{0v}\hat{\tau}_{0v}^{-1} \Big/ \sum_{u,v=0}^{1} r_{uv}\hat{\tau}_{uv}^{-1}, \qquad \widehat{Sp}_2 = \sum_{u=0}^{1} r_{u0}\hat{\tau}_{u0}^{-1} \Big/ \sum_{u,v=0}^{1} r_{uv}\hat{\tau}_{uv}^{-1}.$$

Using information matrix as well as the Delta method, Zhou obtained the estimated variances of $\widehat{Se}_1 - \widehat{Se}_2$ and $\widehat{Sp}_1 - \widehat{Sp}_2$, denoted by $\widehat{\mathrm{Var}}(\widehat{Se}_1 - \widehat{Se}_2)$ and $\widehat{\mathrm{Var}}(\widehat{Sp}_1 - \widehat{Sp}_2)$, respectively. The expressions of $\widehat{\mathrm{Var}}(\widehat{Se}_1 - \widehat{Se}_2)$ and $\widehat{\mathrm{Var}}(\widehat{Sp}_1 - \widehat{Sp}_2)$ are given in Equations (9) and (10), respectively, in Zhou.[8]

Zhou's test statistics for comparing two sensitivities and specificities in the presence of verification bias are

$$X_z^2 = \frac{(\widehat{Se}_1 - \widehat{Se}_2)^2}{\widehat{\mathrm{Var}}(\widehat{Se}_1 - \widehat{Se}_2)}, \qquad G_z^2 = \frac{(\widehat{Sp}_1 - \widehat{Sp}_2)^2}{\widehat{\mathrm{Var}}(\widehat{Sp}_1 - \widehat{Sp}_2)}, \tag{1}$$

Since $\widehat{Se}_k$ and $\widehat{Sp}_k, k = 1, 2$ involve the inverses of all four estimated verification probabilities, $1/\hat{\tau}_{uv}, u, v = 0, 1$, Zhou's test statistics X_z^2 and G_z^2 are not well-defined if any of the four verification probabilities, $\tau_{uv}, u, v = 0, 1$, is zero. As a result, Zhou's test statistics cannot be used in settings where the gold standard test is too invasive or costly to be applied to the subjects who screen negative on both screening tests.

2.2 | Alonzo's test statistics

To overcome the limitation of Zhou's test statistics mentioned in last section, Alonzo[9] proposed alternative test statistics for comparing two sensitivities and specificities in the presence of verification bias. They are constructed based on the ratio of the maximum likelihood estimators of two true positive rates (rTPR) and false positive rates (rFPR). Let $r\widehat{\mathrm{TPR}}$ and $r\widehat{\mathrm{FPR}}$ be the maximum likelihood estimators of rTPR and rFPR, respectively, that is,

$$r\widehat{\mathrm{TPR}} = \frac{s_{11}\hat{\tau}_{11}^{-1} + s_{10}\hat{\tau}_{10}^{-1}}{s_{11}\hat{\tau}_{11}^{-1} + s_{01}\hat{\tau}_{01}^{-1}}, \qquad r\widehat{\mathrm{FPR}} = \frac{r_{11}\hat{\tau}_{11}^{-1} + r_{10}\hat{\tau}_{10}^{-1}}{r_{11}\hat{\tau}_{11}^{-1} + r_{01}\hat{\tau}_{01}^{-1}}.$$

Alonzo has shown that $r\widehat{\text{TPR}}$ and $r\widehat{\text{FPR}}$ can be expressed as a function of six independent binomial proportions, which, in combination with Delta method, leads to the estimated variance formulas for $r\widehat{\text{TPR}}$ and $r\widehat{\text{FPR}}$, given by Equations (7) and (8), respectively, in Alonzo.[9]

Alonzo's test statistics for comparing two sensitivities and specificities in the presence of verification bias are

$$X_a^2 = \frac{[\log(r\widehat{\text{TPR}})]^2}{\widehat{\text{Var}}[\log(r\widehat{\text{TPR}})]}, \qquad G_a^2 = \frac{[\log(r\widehat{\text{FPR}})]^2}{\widehat{\text{Var}}[\log(r\widehat{\text{FPR}})]}. \tag{2}$$

It is important to observe that since $r\widehat{\text{TPR}}$ and $r\widehat{\text{FPR}}$ only involve the inverses of three estimated verification probabilities, $1/\hat{\tau}_{11}, 1/\hat{\tau}_{10}$ and $1/\hat{\tau}_{01}$, Alonzo's test statistics X_a^2 and G_a^2 do not require subjects that screen negative on both tests to receive the gold standard test, and hence it is more flexible and can be used in a sampling scheme where $\tau_{00} = 0$.

3 | PROPOSED METHOD

In this section, we propose a McNemar-type test statistic for comparing two sensitivities (specificities) in the presence of verification bias. Unlike the two reviewed methods, the proposed test statistic only involves the data from discordant pairs and hence it can be used for a sampling scheme where $\tau_{11} = \tau_{00} = 0$. More importantly, the proposed test statistic is more intuitive and much simpler to compute than the two reviewed methods.

When the disease status is verified for all discordant pairs: $(T_{1i} = 1, T_{2i} = 0, D_i = 1)$ and $(T_{1i} = 0, T_{2i} = 1, D_i = 1)$, the traditional McNemar's test statistic can be used to test $H_0 : Se_1 = Se_2$, against the alternative hypothesis $H_a : Se_1 \neq Se_2$, that is,

$$X_m^2 = \frac{(s_{10} - s_{01})^2}{s_{10} + s_{01}}.$$

However, when the disease status is only verified for a subset of discordant pairs, the McNemar's test statistic applied directly to the verified discordant pairs might not be valid as the expected value of $s_{10} - s_{01}$ under the null hypothesis can be nonzero due to the oversampling of discordant pair (1,0) or (0,1).

To address this bias issue, we modify the traditional McNemar's test statistic X_m^2 so that the modified X_m^2 can be used to test for $H_0 : Se_1 = Se_2$ in the presence of verification bias. Since the null hypothesis $H_0 : Se_1 = Se_2$ is equivalent to $H_0 : p_{101} = p_{011}$, a general form of the test statistic for $H_0 : Se_1 = Se_2$, asymptotically distributed as a chi-square with one degree of freedom, is

$$X_{wm}^2 = \frac{(\hat{p}_{101} - \hat{p}_{011})^2}{\widehat{\text{Var}}(\hat{p}_{101} - \hat{p}_{011})|_{H_0}}. \tag{3}$$

We now obtain an expression for X_{wm}^2. Let $\xi_{uv} = \Pr(D = 1|T_1 = u, T_2 = v)$ and $\eta_{uv} = \Pr(T_1 = u, T_2 = v)$ for $u, v = 0, 1$. Zhou[8] showed that the maximum likelihood estimates of ξ_{uv} and η_{uv} are

$$\hat{\xi}_{uv} = \frac{s_{uv}}{m_{uv}}, \qquad \hat{\eta}_{uv} = \frac{n_{uv}}{n} \quad \text{for} \quad u, v = 0, 1.$$

Since $p_{uv0} = (1 - \xi_{uv})\eta_{uv}$ and $p_{uv1} = \xi_{uv}\eta_{uv}$, the maximum likelihood estimates of p_{uv0} and p_{uv1} are

$$\hat{p}_{uv0} = \frac{r_{uv}}{m_{uv}} \frac{n_{uv}}{n}, \qquad \hat{p}_{uv1} = \frac{s_{uv}}{m_{uv}} \frac{n_{uv}}{n} \quad \text{for} \quad u, v = 0, 1, \tag{4}$$

respectively. It is clear that the bias-corrected estimator $\hat{p}_{uv0}$ or $\hat{p}_{uv1}$ is a reweighting estimator,[10] where the observed data are multiplied by the inverse of the estimated disease verification probabilities or sampling fractions, that is, $1/\hat{\tau}_{uv} = n_{uv}/m_{uv}, u, v = 0, 1$.

To obtain an expression for $\widehat{\text{Var}}(\hat{p}_{101} - \hat{p}_{011})|_{H_0}$, we now derive the asymptotic normality result for $\sqrt{n}(\hat{p} - p)$ where $p = p_{101} - p_{011}$ and $\hat{p} = \hat{p}_{101} - \hat{p}_{011}$.

Let $\boldsymbol{\xi} = (\xi_{10}, \xi_{01})^T$ and $\hat{\boldsymbol{\xi}} = (\hat{\xi}_{10}, \hat{\xi}_{01})^T$. In Appendix, we show that $\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi})$ is asymptotically normal with mean 0 and variance-covariance matrix Σ_1, where

$$\Sigma_1 = \begin{bmatrix} \xi_{10}(1-\xi_{10})/(\eta_{10}\tau_{10}) & 0 \\ 0 & \xi_{01}(1-\xi_{01})/(\eta_{01}\tau_{01}) \end{bmatrix}. \tag{5}$$

Similarly, let $\boldsymbol{\eta} = (\eta_{10}, \eta_{01})^T$ and $\hat{\boldsymbol{\eta}} = (\hat{\eta}_{10}, \hat{\eta}_{01})^T$. Then $\sqrt{n}(\hat{\boldsymbol{\eta}} - \boldsymbol{\eta})$ is asymptotically normal with mean 0 and variance-covariance matrix Σ_2, where

$$\Sigma_2 = \mathrm{diag}(\boldsymbol{\eta}) - \boldsymbol{\eta}\boldsymbol{\eta}^T.$$

Since $p = \xi_{10}\eta_{10} - \xi_{01}\eta_{01}$ is a function of $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$, using the Delta method, we can see that $\sqrt{n}(\hat{p} - p)$ is asymptotically normal with mean 0 and variance

$$\sigma_1^2 = \frac{\partial p}{\partial \boldsymbol{\xi}}^T \Sigma_1 \frac{\partial p}{\partial \boldsymbol{\xi}} + \frac{\partial p}{\partial \boldsymbol{\eta}}^T \Sigma_2 \frac{\partial p}{\partial \boldsymbol{\eta}}, \tag{6}$$

where $\frac{\partial p}{\partial \xi} = (\eta_{10}, -\eta_{01})^T$ and $\frac{\partial p}{\partial \boldsymbol{\eta}} = (\xi_{10}, -\xi_{01})^T$. Therefore,

$$\sigma_1^2 = p_{101}\left[\xi_{10} + (1-\xi_{10})/\tau_{10}\right] + p_{011}\left[\xi_{01} + (1-\xi_{01})/\tau_{01}\right] - p^2. \tag{7}$$

Consequently, the asymptotic variance of $\hat{p}$ under the null hypothesis $H_0 : p_{101} = p_{011}$ can be consistently estimated by

$$\widehat{\mathrm{Var}}(\hat{p})|_{H_0} = \frac{1}{n^2}\left\{\left[\hat{\xi}_{10} + (1-\hat{\xi}_{10})/\hat{\tau}_{10}\right]s_{10}/\hat{\tau}_{10} + \left[\hat{\xi}_{01} + (1-\hat{\xi}_{01})/\hat{\tau}_{01}\right]s_{01}/\hat{\tau}_{01}\right\},$$

which, along with (4), leads to the following proposed test statistic for $H_0 : p_{101} = p_{011}$, against the alternative hypothesis $H_a : p_{101} \neq p_{011}$, as

$$\begin{aligned} X_{wm}^2 &= \frac{(\hat{p}_{101} - \hat{p}_{011})^2}{\widehat{\mathrm{Var}}(\hat{p})|_{H_0}} = \frac{(s_{10}/\hat{\tau}_{10} - s_{01}/\hat{\tau}_{01})^2}{[\hat{\xi}_{10} + (1-\hat{\xi}_{10})/\hat{\tau}_{10}]s_{10}/\hat{\tau}_{10} + [\hat{\xi}_{01} + (1-\hat{\xi}_{01})/\hat{\tau}_{01}]s_{01}/\hat{\tau}_{01}} \\ &= \frac{(s_{10}n_{10}/m_{10} - s_{01}n_{01}/m_{01})^2}{[s_{10}/m_{10} + r_{10}n_{10}/m_{10}^2]s_{10}n_{10}/m_{10} + [s_{01}/m_{01} + r_{01}n_{01}/m_{01}^2]s_{01}n_{01}/m_{01}}, \end{aligned} \tag{8}$$

Under the null hypothesis, X_{wm}^2 is asymptotically distributed as a chi-square variable with one degree of freedom.

In a similar manner, we can show that $\sqrt{n}[\hat{p}_{100} - \hat{p}_{010} - (p_{100} - p_{010})]$ is asymptotically normal with mean 0 and variance

$$\sigma_2^2 = p_{100}\left[1 - \xi_{10} + \xi_{10}/\tau_{10}\right] + p_{010}\left[1 - \xi_{01} + \xi_{01}/\tau_{01}\right] - (p_{100} - p_{010})^2.$$

Therefore, the weighted McNemar's test statistic for testing the null hypothesis $H_0 : p_{100} = p_{010}$ against the alternative hypothesis $H_a : p_{101} \neq p_{011}$, is

$$\begin{aligned} G_{wm}^2 &= \frac{(r_{10}/\hat{\tau}_{10} - r_{01}/\hat{\tau}_{01})^2}{[1 - \hat{\xi}_{10} + \hat{\xi}_{10}/\hat{\tau}_{10}]r_{10}/\hat{\tau}_{10} + [\hat{1} - \xi_{01} + \hat{\xi}_{01}/\hat{\tau}_{01}]r_{01}/\hat{\tau}_{01}} \\ &= \frac{(r_{10}n_{10}/m_{10} - r_{01}n_{01}/m_{01})^2}{[r_{10}/m_{10} + s_{10}n_{10}/m_{10}^2]r_{10}n_{10}/m_{10} + [r_{01}/m_{01} + s_{01}n_{01}/m_{01}^2]r_{01}n_{01}/m_{01}}, \end{aligned} \tag{9}$$

which, under the null hypothesis, is asymptotically distributed as a chi-square variable with one degree of freedom.

Since $s_{10}/\hat{\tau}_{10}(r_{10}/\hat{\tau}_{10})$ and $s_{01}/\hat{\tau}_{01}(r_{01}/\hat{\tau}_{01})$ are the weighted counts of (1, 0) pairs and (0, 1) pairs, respectively, the proposed test X_{wm}^2 and G_{wm}^2 can be viewed as the weighted versions of a McNemar's test and it is referred to as the weighted McNemar's tests. Note that when the all discordant pairs are verified, that is, $\tau_{01} = \tau_{10} = 1$, X_{wm}^2 and G_{wm}^2 reduce to the

McNemar test statistics

$$X_m^2 = \frac{(s_{10} - s_{01})^2}{s_{10} + s_{01}} \quad \text{and} \quad G_m^2 = \frac{(r_{10} - r_{01})^2}{r_{10} + r_{01}}, \tag{10}$$

respectively. Therefore, the proposed weighted McNemar's test statistics X_{wm}^2 and G_{wm}^2 are extended versions of a McNemar's test.

4 | MCNEMAR'S TEST STATISTICS IN CASE OF $\tau_{10} = \tau_{01}$

In this section, we consider a special case where the verification probability is the same for both discordant pair of (0,1) and (1,0), that is, $\tau_{01} = \tau_{10} = \tau < 1$. In this case, as will be seen from Equation (12), the McNemar tests X_m^2 and G_m^2 can be applied directly to the verified study subjects without incurring any bias. However, we show that they are less powerful than their counterparts of the proposed weighted McNemar tests X_{wm}^2 and G_{wm}^2.

To proceed, we now consider estimating p_{101} and p_{011} using the known value of τ, $\tilde{p}_{101} = s_{10}/(n\tau)$ and $\tilde{p}_{011} = s_{01}/(n\tau)$. Let $\tilde{p} = \tilde{p}_{101} - \tilde{p}_{011}$. Note that $s_{10} = \sum_{i=1}^{n} R_i D_i T_{1i}(1 - T_{2i})$ and $s_{01} = \sum_{i=1}^{n} R_i D_i (1 - T_{1i}) T_{2i}$. Therefore, $\tilde{p}$ can be expressed as an independent sum:

$$\tilde{p} = \frac{s_{10}}{n\tau} - \frac{s_{01}}{n\tau} = \frac{1}{n\tau} \sum_{i=1}^{n} R_i D_i (T_{1i} - T_{2i}), \tag{11}$$

Therefore, $\sqrt{n}(\tilde{p} - p)$ is asymptotically normal with mean 0 and variance

$$\sigma_3^2 = \mathrm{Var}\left[\frac{1}{\tau} R_i D_i (T_{1i} - T_{2i})\right] = \frac{p_{101} + p_{011}}{\tau} - p^2.$$

Consequently, the asymptotic variance of $\tilde{p}$ under the null hypothesis $H_0 : p_{101} = p_{011}$ can be consistently estimated by

$$\widehat{\mathrm{Var}}(\tilde{p})|_{H_0} = \frac{s_{10} + s_{01}}{n^2 \tau^2},$$

and hence the McNemar's test statistic X_m^2 can be expressed as

$$X_m^2 = \frac{(s_{10} - s_{01})^2}{s_{10} + s_{01}} = \frac{\tilde{p}^2}{\widehat{\mathrm{Var}}(\tilde{p})|_{H_0}}. \tag{12}$$

Comparing σ_3^2 with σ_1^2 in (5) with τ_{10} and τ_{01} replaced by τ, we can see that $\sigma_3^2 > \sigma_1^2$ and hence X_m^2 is less powerful than the proposed weighted McNemar tests X_{wm}^2. In the similar manner, we can show that G_m^2 is less powerful than the proposed weighted McNemar tests G_{wm}^2.

5 | WEIGHTED MCNEMAR'S TEST STATISTIC WITH DISCRETE COVARIATES

Assume that the vector of discrete covariates, $\boldsymbol{X} = (X_1, \dots, X_k)$, has a number J of different covariate patterns.[11] Let $x_j = (x_{j1}, \cdot, x_{jk})$ denote the jth covariate pattern of k covariates, $j = 1, \dots, J$. Assume that the probability of selection of a subject for disease verification depends on not only the results of the screening tests but also observed discrete covariates $\boldsymbol{X}$. In this section, we extend the weighted McNemar's test in Section 3 to a more general case with discrete covariates.

Let T_{1i}, T_{2i}, and $\boldsymbol{X}_i$ denote the observed values of T_1, T_2 and $\boldsymbol{X}$ from the ith subject, $i = 1, \dots, n$. For the jth stratum, we use n_{uvj} and m_{uvj} to denote the numbers of subjects and verified subjects with $(T_1 = u, T_2 = v, \boldsymbol{X} = x_j)$, respectively. Similarly, we use r_{uvj} and s_{uvj} to denote the numbers of verified subjects with $(T_1 = u, T_2 = v, \boldsymbol{X} = x_j, D = 0)$ and $(T_1 = u, T_2 = v, \boldsymbol{X} = x_j, D = 1)$, respectively, $u, v = 0, 1$.

Let $\xi_{uvj} = \Pr(D = 1|T_1 = u, T_2 = v, \boldsymbol{X} = x_j)$ and $\eta_{uvj} = \Pr(T_1 = u, T_2 = v, \boldsymbol{X} = x_j)$. Then $p = p_{101} - p_{011}$ can be expressed in terms of ξ_{uvj}s and η_{uvj}s as

$$p = \sum_{j=1}^{J} (\xi_{10j}\eta_{10j} - \xi_{01j}\eta_{01j}). \tag{13}$$

Since the maximum likelihood estimators of ξ_{uvj} and η_{uvj} are $\hat{\xi}_{uvj} = s_{uvj}/m_{uvj}$ and $\hat{\eta}_{uvj} = n_{uvj}/n$, respectively. The maximum likelihood estimator of p can be obtained by substituting ξ_{uvj} and η_{uvj} in (13) by their corresponding maximum likelihood estimators, that is,

$$\hat{p} = \frac{1}{n}\sum_{j=1}^{J} (s_{10j}\hat{\tau}_{10j} - s_{01j}\hat{\tau}_{01j}),$$

where $\hat{\tau}_{uvj} = m_{uvj}/n_{uvj}$.

Let $\boldsymbol{\beta} = (\xi_{10j}, \xi_{01j}, j = 1, \ldots, J)^T$ and $\hat{\boldsymbol{\beta}} = (\hat{\xi}_{10j}, \hat{\xi}_{01j}, j = 1, \ldots, J)^T$. Then $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ is asymptotically normal with mean 0 and variance-covariance matrix Ω_1, where Ω_1 is a $2J \times 2J$ diagonal matrix with diagonal elements $\xi_{10j}(1 - \xi_{10j})/(\eta_{10j}\tau_{10j})$, $\xi_{01j}(1 - \xi_{01j})/(\eta_{01j}\tau_{01j}), j = 1, \ldots, J$. Similarly, let $\boldsymbol{\gamma} = (\eta_{10j}, \eta_{01j}, j = 1, \ldots, J)^T$ and $\hat{\boldsymbol{\gamma}} = (\hat{\eta}_{10j}, \hat{\eta}_{01j}, j = 1, \ldots, J)^T$. Then $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})$ is asymptotically normal with mean 0 and variance-covariance matrix Ω_2, where Ω_2 is given by

$$\Omega_2 = \text{diag}(\boldsymbol{\gamma}) - \boldsymbol{\gamma}\boldsymbol{\gamma}^T.$$

Again, since p is a function of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, using the Delta method, we can see that $\sqrt{n}(\hat{p} - p)$ is asymptotically normal with mean 0 and variance

$$\begin{aligned}
\sigma_p^2 &= \frac{\partial p}{\partial \boldsymbol{\beta}}^T \Omega_1 \frac{\partial p}{\partial \boldsymbol{\beta}} + \frac{\partial p}{\partial \boldsymbol{\gamma}}^T \Omega_2 \frac{\partial p}{\partial \boldsymbol{\gamma}} \\
&= \sum_{j=1}^{J} \eta_{10}\xi_{10j}\left[\xi_{10j} + \frac{1 - \xi_{10j}}{\tau_{10j}}\right] + \sum_{j=1}^{J} \eta_{01j}\xi_{01j}\left[\xi_{01j} + \frac{1 - \xi_{01j}}{\tau_{01j}}\right] - p^2.
\end{aligned}$$

Therefore, the asymptotic variance of $\hat{p}$ under the null hypothesis $H_0 : p_{101} = p_{011}$ can be consistently estimated by

$$\widehat{Var}(\hat{p})|_{H_0} = \frac{1}{n^2}\left\{\sum_{j=1}^{J}\left[\hat{\xi}_{10j} + \frac{1 - \hat{\xi}_{10j}}{\hat{\tau}_{10j}}\right] s_{10j}/\hat{\tau}_{10j} + \sum_{j=1}^{J}\left[\hat{\xi}_{01j} + \frac{1 - \hat{\xi}_{01j}}{\hat{\tau}_{01j}}\right] s_{01j}/\hat{\tau}_{01j}\right\}.$$

In the presence of covariates, the proposed test statistic for testing the null hypothesis $H_0 : p_{101} = p_{011}$ against the alternative hypothesis $H_a : p_{101} \neq p_{011}$, is

$$X_{wmc}^2 = \frac{\left\{\sum_{j=1}^{J}\left(s_{10j}n_{10j}/m_{10j} - s_{01j}n_{01j}/m_{01j}\right)\right\}^2}{\sum_{j=1}^{J}\left\{[s_{10j}/m_{10j} + r_{10j}n_{10j}/m_{10j}^2]s_{10j}n_{10j}/m_{10j} + [s_{01j}/m_{01j} + r_{01j}n_{01j}/m_{01j}^2]s_{01j}n_{01j}/m_{01j}\right\}}, \tag{14}$$

which, under the null hypothesis, is asymptotically distributed as a chi-square variable with one degree of freedom.

Similarly, the proposed test statistic for testing the null hypothesis $H_0 : p_{100} = p_{010}$ against the alternative hypothesis $H_a : p_{101} \neq p_{011}$ in the presence of covariates, is

$$G_{wmc}^2 = \frac{\left\{\sum_{j=1}^{J}\left(r_{10j}n_{10j}/m_{10j} - r_{01j}n_{01j}/m_{01j}\right)\right\}^2}{\sum_{j=1}^{J}\left\{[r_{10j}/m_{10j} + s_{10j}n_{10j}/m_{10j}^2]r_{10j}n_{10j}/m_{10j} + [r_{01j}/m_{01j} + s_{01j}n_{01j}/m_{01j}^2]r_{01j}n_{01j}/m_{01j}\right\}}, \tag{15}$$

which, under the null hypothesis, is asymptotically distributed as a chi-square variable with one degree of freedom.

6 SIMULATION

We conducted a Monte Carlo simulation study to assess the size and power of the proposed weighted McNemar's test statistics X^2_{wm} and G^2_{wm} in (8) and (9) and to compare them with their counterparts: McNemar's test statistics X^2_m and G^2_m in (10), Zhou's test statistics X^2_z and G^2_z in (1), and Alonzo's test statistics X^2_a and G^2_a in (2).

We generate the full data $(D_i, T_{1i}, T_{2i}), i = 1, \ldots, n$ as follows. We first describe a model for the joint distribution for (D, T_1, T_2). Let $q_{ij} = \Pr(T_1 = i, T_2 = j|D = 1), i,j = 0, 1$ and ρ_1 be the correlation coefficient of T_1 and T_2 conditional on $D = 1$. Then

$$\rho_1 = \frac{q_{11} - Se_1Se_2}{\sqrt{Se_1(1 - Se_1)Se_2(1 - Se_2)}}.$$

Since $q_{11} + q_{10} = Se_1$, $q_{11} + q_{01} = Se_2$, $q_{11} = Se_1Se_2 + \rho_1\{Se_1(1 - Se_1)Se_2(1 - Se_2)\}^{-1/2}$, the multinominal probability q_{ij} can be expressed in terms of Se_1, Se_2, and ρ_1 as

$$q_{ij} = Se_1^i(1 - Se_1)^{1-i}Se_2^j(1 - Se_2)^{1-j}(1 + \rho_1 v_1 v_2), i,j = 0, 1,$$

where $v_1 = \{i - Se_1\}\{Se_1(1 - Se_1)\}^{-1/2}$ and $v_2 = \{j - Se_2\}\{Se_2(1 - Se_2)\}^{-1/2}$.

Similarly, let $p_{ij} = \Pr(T_1 = i, T_2 = j|D = 0), i,j = 0, 1$ and ρ_0 be the correlation coefficient of T_1 and T_2 conditional on $D = 0$. Then the multinominal probability p_{ij} can be expressed in terms of Sp_1, Sp_2, and ρ_0 as

$$p_{ij} = (1 - Sp_1)^i Sp_1^{1-i}(1 - Sp_2)^j Sp_2^{1-j}(1 + \rho_0 u_1 u_2), i,j = 0, 1,$$

where $u_1 = \{i - (1 - Sp_1)\}\{Sp_1(1 - Sp_1)\}^{-1/2}$ and $u_2 = \{j - (1 - Sp_2)\}\{Sp_2(1 - Sp_2)\}^{-1/2}$. Therefore, the full data $(D_i, T_{1i}, T_{2i}), i = 1, \ldots, n$ can be generated from the multinominal distribution of (D, T_1, T_2) defined by $\Pr(T_1 = u, T_2 = v, D = w) = wq_{uv}\lambda + (1 - w)p_{uv}(1 - \lambda)$, which is a function of the parameters, $\lambda, Se_1, Se_2, Sp_1, Sp_2, \rho_0$ and ρ_1. Finally, we generate the verification indicator R_i from the Bernoulli distribution with the probability

$$\Pr(R_i = 1|T_{1i}, T_{2i}) = \tau_{11}T_{1i}T_{2i} + \tau_{10}T_{1i}(1 - T_{2i}) + \tau_{01}(1 - T_{1i})T_{2i} + \tau_{00}(1 - T_{1i})(1 - T_{2i}).$$

The parameters, τ_{11}, τ_{10}, τ_{01}, τ_{00}, λ, Se_1, Se_2, Sp_1, Sp_2, ρ_0, and ρ_1, were used to formulate a simulation scenario. For each considered true scenario, we generated data 4 million times because results are displayed to three decimal digits, and this number of simulations provides the length of the 95% confidence interval for the proportion of rejected tests to be less than 0.001. We added a very small number, say 0.0001, to each cell of generated data when this particular dataset led to inability to compute one of the considered statistics because some of $\hat{\tau}_{ij} = n_{ij}/m_{ij}, i.j = 0, 1$ can not be calculated as a result of $m_{ij} = 0$. For each simulation, H_0 was rejected if a test statistic exceeded the 0.95 quantile of the chi-square distribution with one degree of freedom.

Simulation results are summarized in Tables 2 and 3, which indicate the case where a test statistic cannot be calculated because it is not well-defined. As seen in Tables 2 and 3, the empirical size and power of Zhou's test statistics are not available when any of the two verification probabilities for accordant pairs, τ_{00} and τ_{11}, is zero. The unavailability of empirical size and power is also true with Alonzo's test statistics when $\tau_{11} = 0$. On the other hand, however, the empirical size and power of the weighted McNemar's test statistics are available as long as the two verification probabilities for discordant pairs, τ_{01} and τ_{10}, are positive.

Table 2 displays empirical size for weighted McNemar's statistic and its counterparts: McNemar's statistic, Alonzo's statistic, and Zhou's statistic. We considered $\lambda = 0.3$, $\rho_0 = \rho_1 = 0.3$, $Se_1 = Se_2 = 0.8$ and $Sp_1 = Sp_2 = 0.9$ and varied the verification probabilities and total sample size n. In assessing the size of the McNemar's test applied to the verified subjects, we observe that the McNemar's test preserves the nominal 0.05 type I error well when the verification probabilities for the discordant pairs are equal ($\tau_{10} = \tau_{01} = 0.5$). However, when the verification probabilities for the discordant pairs are not equal ($\tau_{10} = 0.5, \tau_{01} = 0.9$), the empirical size of the McNemar's test exceeded the nominal level, as anticipated from the results given in Section 4. In the case of all four positive verification probabilities so that all three test statistics, Zhou's test statistic, Alonzo's test statistic, and the proposed weighted McNemar's test statistic, can be calculated, we observe that the proposed weighted McNemar's test statistic preserves the nominal 0.05 type I error better or equally well to Zhou's test statistic and Alonzo's test statistic.

TABLE 2 Empirical sizes (%) of McNemar's statistics (X_m^2, G_m^2), Zhou's statistics (X_z^2, G_z^2), Alonzo's statistics (X_a^2, G_a^2) and proposed weighted McNemar's statistics (X_{wm}^2, G_{wm}^2). True sensitivities and specificities are $Se_1 = Se_2 = 0.8, Sp_1 = Sp_2 = 0.9$. "—" indicates the case where a test statistic cannot be calculated as it is not well-defined

						$H_0: Se_1 = Se_2$				$H_0: Sp_1 = Sp_2$			
τ_{00}	τ_{11}	τ_{10}	τ_{01}	n	n^*	X_m^2	X_z^2	X_a^2	X_{wm}^2	G_m^2	G_z^2	G_a^2	G_{wm}^2
0.0	0.0	0.5	0.5	50	4	0.9	—	—	6.4	1.8	—	—	7.7
				100	8	3.5	—	—	9.0	4.4	—	—	9.5
				300	24	4.4	—	—	6.0	4.4	—	—	5.6
				500	39	4.8	—	—	5.4	5.3	—	—	5.2
				1000	78	5.0	—	—	5.2	4.8	—	—	5.0
		0.9	0.5	50	6	3.1	—	—	4.7	5.3	—	—	6.1
				100	11	8.4	—	—	8.9	1—	—	—	8.6
				300	33	18.1	—	—	5.9	23.7	—	—	5.6
				500	55	29.3	—	—	5.4	36.0	—	—	5.2
				1000	109	50.4	—	—	5.2	61.9	—	—	5.1
0.0	0.8	0.5	0.5	50	14	0.9	—	6.3	6.5	1.7	—	1.9	7.7
				100	27	3.5	—	9.7	9.0	4.5	—	6.5	9.6
				300	80	4.2	—	6.0	5.9	4.6	—	4.8	5.8
				500	132	5.0	—	5.6	5.6	5.3	—	4.8	5.3
				1000	264	5.0	—	5.2	5.2	4.9	—	4.9	5.2
		0.9	0.5	50	15	3.2	—	4.6	4.8	5.3	—	1.4	6.2
				100	30	8.4	—	9.1	9.0	9.9	—	5.3	8.6
				300	89	17.8	—	5.8	5.8	23.3	—	4.7	5.5
				500	148	29.4	—	5.3	5.3	36.3	—	4.8	5.3
				1000	295	5	—	5.1	5.1	61.5	—	4.9	5.1
0.2	0.0	0.5	0.5	50	11	0.9	—	—	6.5	1.7	—	—	7.6
				100	21	3.5	—	—	9.2	4.4	—	—	9.6
				300	61	4.4	—	—	6.0	4.6	—	—	5.7
				500	101	5.0	—	—	5.5	5.3	—	—	5.3
				1000	201	4.9	—	—	5.2	5.0	—	—	5.2
		0.9	0.5	50	12	3.2	—	—	4.8	5.3	—	—	6.2
				100	24	8.5	—	—	9.0	9.9	—	—	8.6
				300	70	17.7	—	—	5.7	23.3	—	—	5.5
				500	116	29.2	—	—	5.3	36.0	—	—	5.2
				1000	232	50.2	—	—	5.1	61.6	—	—	5.2
0.2	0.4	0.5	0.5	50	15	0.9	9.1	5.3	6.5	1.7	8.1	0.4	7.6
				100	30	3.6	11.8	8.8	8.9	4.4	9.9	1.7	9.4
				300	88	4.3	6.6	6.0	6.0	4.5	5.9	3.6	5.7
				500	147	4.8	5.8	5.5	5.5	5.3	5.5	4.2	5.4
				1000	293	5.1	5.4	5.3	5.3	4.9	5.2	4.6	5.1

(Continues)

TABLE 2 (Continued)

						$H_0: Se_1 = Se_2$				$H_0: Sp_1 = Sp_2$			
τ_{00}	τ_{11}	τ_{10}	τ_{01}	n	n^*	X^2_m	X^2_z	X^2_a	X^2_{wm}	G^2_m	G^2_z	G^2_a	G^2_{wm}
		0.9	0.5	50	17	3.2	7.2	3.9	4.9	5.1	6.7	0.3	6.0
				100	33	8.3	10.2	8.3	8.9	10.0	9.3	1.6	8.8
				300	98	17.7	6.2	5.8	5.8	23.2	5.8	3.8	5.6
				500	163	28.9	5.5	5.3	5.3	36.0	5.3	4.2	5.2
				1000	325	50.3	5.3	5.2	5.2	61.6	5.2	4.7	5.1
0.2	0.8	0.5	0.5	50	20	1.0	9.5	6.4	6.6	1.8	8.2	1.8	7.6
				100	39	3.6	12.1	9.6	9.0	4.5	10.2	6.5	9.6
				300	116	4.4	6.8	6.3	6.1	4.5	5.8	4.6	5.5
				500	193	4.9	5.7	5.4	5.3	5.3	5.5	4.8	5.4
				1000	386	5.0	5.3	5.2	5.1	5.1	5.4	5.1	5.4
		0.9	0.5	50	21	3.2	7.4	4.6	4.8	5.2	6.8	1.3	6.1
				100	42	8.3	10.4	9.0	8.9	9.9	9.3	5.4	8.8
				300	126	17.8	6.3	5.9	5.9	23.6	5.7	4.7	5.5
				500	209	29.3	5.6	5.5	5.5	36.3	5.4	4.8	5.3
				1000	418	50.0	5.3	5.2	5.2	61.8	5.1	4.9	5.1

Table 3 summarizes empirical powers when the true sensitivities and specificities are $Se_1 = 0.8, Se_2 = 0.75, Sp_1 = 0.9$ and $Sp_2 = 0.85$. Here we also consider $\lambda = 0.3$, $\rho_0 = \rho_1 = 0.3$, and vary the verification probabilities and total sample size n. It can be seen that in the case of equal discordant pair verification probabilities, $\tau_{10} = \tau_{01}$, the empirical power of the proposed weighted McNemar's test statistic X^2_{wm} is higher than its counterpart of McNemar's test statistic X^2_m, for example, when $\tau_{10} = \tau_{01} = 0.5$ and $n = 1000$, the empirical power of X^2_{wm} is 23.5 while the empirical power of X^2_m is 29.4. This is also true with the empirical power comparison between G^2_{wm} and G^2_m. In the case where all four verification probabilities are positive, the empirical power of the proposed weighted McNemar's test statistic is slightly higher or similar to the powers of Zhou's and Alonzo's test statistics. The empirical power of Alonzo' test statistic may be somewhat underestimated because of the deflated type I error. To see how much Zhou's test and Alonzo's test would benefit from verifying more subjects with T_1 and T_2 being positive, two values of τ_{11}, 0.4 and 0.8, were considered. It can be seen that Zhou's test and Alonzo's test benefit little from the increasing value of τ_{11}. For example, when $\tau_{11} = 0.4(\tau_{00} = 0.2, \tau_{10} = 0.9, \tau_{01} = 0.5, n = 300)$, the empirical power of X^2_z is 14.7 while when $\tau_{11} = 0.8(\tau_{00} = 0.2, \tau_{10} = 0.9, \tau_{01} = 0.5, n = 300)$, the empirical power of X^2_z is 14.6. This is not surprising because it is well-known that accordant pairs are not informative for comparing two correlated proportions.

In conclusion, the simulation results support use of the proposed weighted McNemar's test statistics due to their good size and power behavior.

7 | DATA EXAMPLE

We apply the results obtained in Sections 3 and 5 to the epidemiology study of Alzheimer's disease[3] described in introduction section. The data is displayed as frequencies in Table 4, where D represents dementia status and T_1 and T_2 represent the new screening test and the standard test, respectively. The same data were also used by Zhou.[8]

7.1 | Analysis of dementia data with subjects aged 75 years or more

We use the dementia data with subjects aged 75 years or more to illustrate the calculations of X^2_{wm} and G^2_{wm}, which may be easily followed by practitioners with their own data. First, we compute the weighted McNemar's test statistic X^2_{wm}

TABLE 3 Empirical powers (%) of McNemar's statistics (X^2_m, G^2_m), Zhou's statistics (X^2_z, G^2_z), Alonzo's statistics (X^2_a, G^2_a), and proposed weighted McNemar's statistics (X^2_{wm}, G^2_{wm}). True sensitivities and specificities are $Se_1 = 0.75$, $Se_2 = 0.8$, $Sp_1 = 0.85$, $Sp_2 = 0.9$. "—" indicates the case where a test statistic cannot be calculated as it is not well-defined

						$H_0 : Se_1 = Se_2$				$H_0 : Sp_1 = Sp_2$			
τ_{00}	τ_{11}	τ_{10}	τ_{01}	n	n^*	X^2_m	X^2_z	X^2_a	X^2_{wm}	G^2_m	G^2_z	G^2_a	G^2_{wm}
0.0	0.0	0.5	0.5	50	5	1.5	—	—	8.6	4.7	—	—	13.5
				100	10	5.3	—	—	12.0	10.6	—	—	19.3
				300	28	9.4	—	—	13.4	24.8	—	—	35.1
				500	46	14.0	—	—	17.6	40.2	—	—	52.8
				1000	92	23.5	—	—	29.4	66.5	—	—	81.8
		0.9	0.5	50	7	7.0	—	—	7.8	3.8	—	—	11.2
				100	13	17.5	—	—	14.9	4.6	—	—	16.6
				300	38	46.9	—	—	15.6	5.5	—	—	38.5
				500	62	70.4	—	—	20.9	5.7	—	—	58.5
				1000	124	93.9	—	—	34.2	6.2	—	—	86.9
0.2	0.0	0.5	0.5	50	11	1.5	—	—	9.1	4.8	—	—	13.7
				100	21	5.3	—	—	12.0	10.6	—	—	19.1
				300	63	9.3	—	—	13.5	25.1	—	—	35.3
				500	105	13.9	—	—	17.4	39.9	—	—	52.7
				1000	210	23.4	—	—	29.3	66.8	—	—	81.9
		0.9	0.5	50	13	7.0	—	—	7.6	3.7	—	—	10.8
				100	25	17.6	—	—	15.1	4.7	—	—	16.8
				300	73	47.3	—	—	15.6	5.8	—	—	38.5
				500	121	70.6	—	—	20.8	5.7	—	—	57.9
				1000	242	93.8	—	—	34.2	6.2	—	—	86.9
0.0	0.8	0.5	0.5	50	14	1.5	—	8.4	8.7	4.7	—	5.6	13.8
				100	28	5.3	—	12.8	12.1	10.5	—	15.0	19.0
				300	83	9.0	—	13.4	13.3	25.1	—	32.6	35.2
				500	138	13.9	—	17.5	17.6	40.0	—	51.3	52.9
				1000	275	23.5	—	29.6	29.6	66.7	—	81.3	81.9
		0.9	0.5	50	16	7.1	—	7.2	7.7	3.8	—	4.8	11.1
				100	31	17.6	—	14.9	15.0	4.7	—	14.4	16.6
				300	93	47.4	—	15.0	15.5	5.8	—	39.1	38.4
				500	154	70.6	—	20.4	20.9	5.6	—	58.9	58.0
				1000	307	94.0	—	33.5	34.0	6.2	—	87.4	86.9
0.2	0.4	0.5	0.5	50	16	1.5	11.3	7.0	8.8	4.6	15.0	1.5	13.6
				100	31	5.4	14.5	11.8	12.1	10.3	20.7	7.1	19.0
				300	91	9.0	13.3	13.1	13.2	25.1	37.0	29.9	35.3
				500	151	14.2	17.5	17.7	17.8	40.0	54.0	49.3	52.7
				1000	301	23.2	28.9	29.3	29.4	66.5	82.3	80.5	81.6

(Continues)

TABLE 3 (Continued)

τ_{00}	τ_{11}	τ_{10}	τ_{01}	n	n^*	$H_0: Se_1 = Se_2$ X_m^2	X_z^2	X_a^2	X_{wm}^2	$H_0: Sp_1 = Sp_2$ G_m^2	G_z^2	G_a^2	G_{wm}^2
		0.9	0.5	50	17	7.0	9.8	5.6	7.6	3.7	13.4	1.6	10.9
				100	34	17.6	15.4	13.6	15.0	4.7	19.0	8.8	16.7
				300	100	47.2	14.6	14.8	15.5	5.5	40.4	36.6	38.5
				500	167	70.7	19.6	20.0	20.9	5.6	59.5	57.5	58.0
				1000	334	93.8	32.8	33.6	34.3	6.3	87.3	86.9	86.8
0.2	0.8	0.5	0.5	50	20	1.5	12.0	8.5	8.8	4.7	15.2	5.6	13.6
				100	40	5.3	14.6	12.5	11.9	10.4	20.7	15.1	18.9
				300	118	9.1	13.4	13.4	13.3	25.3	37.2	33.0	35.5
				500	196	14.2	17.6	17.8	17.8	40.0	54.0	50.9	52.6
				1000	392	23.3	29.0	29.4	29.4	66.7	82.2	81.0	81.6
		0.9	0.5	50	22	7.2	10.5	7.2	7.8	3.7	13.6	4.9	11.0
				100	43	17.3	15.6	14.8	14.9	4.7	19.0	14.5	16.7
				300	128	47.2	14.7	15.0	15.5	5.6	40.2	39.0	38.3
				500	213	70.3	19.6	20.2	20.7	5.7	59.7	59.2	58.3
				1000	425	94.0	32.8	33.6	34.2	6.3	87.3	87.3	86.8

TABLE 4 Data from the dementia study comparing the accuracy of two screening tests for detecting dementia

	$T_1 = 1$		$T_1 = 0$	
	$T_2 = 1$	$T_2 = 0$	$T_2 = 1$	$T_2 = 0$
Age ≥ 75 years				
$R = 1$				
$D = 1$	31	5	3	1
$D = 0$	25	10	19	55
$R = 0$	22	6	65	346
Total	78	21	87	402
Age < 75 years				
$R = 1$				
$D = 1$	7	0	0	0
$D = 0$	10	19	6	34
$R = 0$	9	11	52	759
Total	26	30	58	793

based on the data with subjects aged 75 years or more. The calculation of this statistic requires the numbers of discordant pairs, $n_{10} = 21$, $n_{01} = 87$, the numbers of verified discordant pairs, $m_{10} = 15$, $m_{01} = 22$, the numbers of discordant pairs among verified cases, $s_{10} = 5$, $s_{01} = 3$, and the numbers of discordant pairs among verified controls, $r_{10} = 10$, $r_{01} = 19$. Substituting these four sets of counts into (8) , we obtain

$$X_{wm}^2 = \frac{(s_{10}n_{10}/m_{10} - s_{01}n_{01}/m_{01})^2}{[s_{10}/m_{10} + r_{10}n_{10}/m_{10}^2]s_{10}n_{10}/m_{10} + [s_{01}/m_{01} + r_{01}n_{01}/m_{01}^2]s_{01}n_{01}/m_{01}}$$

$$= \frac{\left(5 \times \frac{21}{15} - 3 \times \frac{87}{22}\right)^2}{\left[\frac{5}{15} + \frac{10\times21}{15^2}\right] \times 5 \times \frac{21}{15} + \left[\frac{3}{22} + \frac{19\times87}{22^2}\right] \times 3 \times \frac{87}{22}} = 0.464.$$

Hence, utilizing χ_1^2 distribution, we have a P-value of .496 and conclude that there is no evidence to claim difference in sensitivity between the new and standard screening tests.

The weighted McNemar's test statistic G_{wm}^2 based on the data with subjects aged 75 years or more is computed by substituting the four sets of counts into (9) as

$$G_{wm}^2 = \frac{(r_{10}n_{10}/m_{10} - r_{01}n_{01}/m_{01})^2}{[r_{10}/m_{10} + s_{10}n_{10}/m_{10}^2]r_{10}n_{10}/m_{10} + [r_{01}/m_{01} + s_{01}n_{01}/m_{01}^2]r_{01}n_{01}/m_{01}}$$
$$= \frac{\left(10 \times \frac{21}{15} - 19 \times \frac{87}{22}\right)^2}{\left[\frac{10}{15} + \frac{5\times21}{15^2})\right] \times 10 \times \frac{21}{15} + \left[\frac{19}{22} + \frac{3\times87}{22^2}\right] \times 19 \times \frac{87}{22}} = 30.82,$$

which yields a P-value $< .001$. Thus, we conclude that the specificity of the new screening test differs from that of the standard screening test.

For comparison, we calculated Zhou's test statistics, X_z^2 and G_z^2, as 0.51 and 34.57, respectively, and Alzno's test statistics X_a^2 and G_a^2, as 0.49 and 31.58, respectively, which lead to the same conclusions as those of the proposed weighted McNemar's test statistics.

7.2 | Analysis of dementia data with all observations

According to the study design, the probability of selecting a subject for a clinical assessment depends on not only the two test results but also a binary covariate, which is equal to one if a subject is 75 years old or older, and zero otherwise. We use the dementia data with all observations to illustrate the calculations of X_{wmc}^2 and G_{wmc}^2.

We first compute the weighted McNemar's test statistic X_{wmc}^2. The square root of the numerator of this statistic is the sum of the square root of the numerators of X_{wm}^2 applied to the two covariate strata. The denominator of this statistic can be calculated by adding the denominators of X_{wm}^2 applied to the two covariate strata. Specifically, the calculation of this statistic requires the numbers of discordant pairs, $n_{101} = 21$, $n_{011} = 87$, $n_{102} = 30$, $n_{012} = 58$, the numbers of verified discordant pairs, $m_{101} = 15$, $m_{011} = 22$, $m_{102} = 19$, $m_{012} = 6$, the numbers of discordant pairs among verified cases, $s_{101} = 5$, $s_{011} = 3$, $s_{102} = 0$, $s_{012} = 0$, and the numbers of discordant pairs among verified controls, $r_{101} = 10$, $r_{011} = 19$, $r_{102} = 19$, $r_{012} = 6$. Substituting these four sets of counts into (14) , we obtain

$$X_{wmc}^2 = \frac{\left(5 \times \frac{21}{15} - 3 \times \frac{87}{22} + 0 \times \frac{30}{19} - 0 \times \frac{58}{6}\right)^2}{\left[\frac{5}{15} + \frac{10\times21}{15^2}\right] \times 5 \times \frac{21}{15} + \left[\frac{3}{22} + \frac{19\times87}{22^2}\right] \times 3 \times \frac{87}{22} + \left[\frac{0}{19} + \frac{19\times30}{19^2}\right] \times 0 \times \frac{30}{19} + \left[\frac{0}{6} + \frac{6\times58}{6^2}\right] \times 0 \times \frac{58}{6}}$$
$$= 0.464.$$

Hence, utilizing χ_1^2 distribution, we have a P-value of .496 and conclude that there is no evidence to claim difference in sensitivity between the new and standard screening tests.

The weighted McNemar's test statistic G_{wmc}^2 is computed by substituting the four sets of counts into (15) as

$$G_{wmc}^2 = \frac{\left(10 \times \frac{21}{15} - 19 \times \frac{87}{22} + 19 \times \frac{30}{19} - 6 \times \frac{58}{6}\right)^2}{\left[\frac{10}{15} + \frac{5\times21}{15^2}\right] \times 10 \times \frac{21}{15} + \left[\frac{19}{22} + \frac{3\times87}{22^2}\right] \times 19 \times \frac{87}{22} + \left[\frac{19}{19} + \frac{0\times30}{19^2}\right] \times 19 \times \frac{30}{19} + \left[\frac{6}{6} + \frac{0\times58}{6^2}\right] \times 6 \times \frac{58}{6}}$$
$$= 37.97.$$

Hence, utilizing χ_1^2 distribution, we have a P-value $< .00001$. Thus, we conclude that the specificity of the new screening test differs from that of the standard screening test.

For comparison, we also calculated Zhou's test statistics in the presence of covariates, X^2_{zc} and G^2_{zc}, as 2.27 and 54.13, respectively, which lead to the same conclusions as those of their respective counterparts of X^2_{wmc} and G^2_{wmc}.

8 | DISCUSSION

We have proposed new weighted McNemar's statistics (8) and (9) for testing equality of sensitivity and specificity of two screening tests in the presence of verification bias. Simulations indicate that these statistics preserve type I error equally well or better than the Zhou's statistic based on the difference of the bias-corrected sensitivities (specificities) or Alonzo's statistics based on the ratio of the bias-corrected sensitivities(specificities). The proposed statistics are intuitive and simple to compute, only involving some minor modification of a McNemar's test statistic using the estimated verification probabilities for discordant pairs. Furthermore, the proposed weighted McNemar's statistics are advantageous over existing methods in that they only require that two verification probabilities be positive, as opposed to Zhou's statistics, which require that all four verification probabilities be positive, and Alonzo's statistics, which require that three verification probabilities be positive. Finally, in the special case that the verification probabilities for the two discordant pairs are equal, the proposed weighted McNemar's statistics are more powerful than McNemar's statistics even though McNemar's statistics preserve type I error as well as the proposed weighted McNemar's statistics in this special case. Thus, we recommend the weighted McNemar's statistic for hypothesis testing over existing methods.

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APPENDIX A. DERIVATION OF EQUATION (5)

First note that $\hat{\xi}_{10} - \xi_{10}$ can be written as

$$\hat{\xi}_{10} - \xi_{10} = \frac{(1/n)\sum_{i=1}^{n} R_i T_{1i}(1 - T_{2i})(D_i - \xi_{10})}{(1/n)\sum_{i=1}^{n} R_i T_{1i}(1 - T_{2i})}.$$

Since denominator converges to $\eta_{10}\tau_{10}$ in probability and the numerator converges in distribution, by the Slutsky Theorem,[12] $\hat{\xi}_{10} - \xi_{10}$ is asymptotically equivalent to the following independent sum:

$$\hat{\xi}_{10} - \xi_{10} = \frac{1}{n\eta_{10}\tau_{10}} \sum_{i=1}^{n} R_i T_{1i}(1 - T_{2i})(D_i - \xi_{10}).$$

Similarly, we can show that $\hat{\xi}_{01} - \xi_{01}$ is asymptotically equivalent to the following independent sum:

$$\hat{\xi}_{01} - \xi_{01} = \frac{1}{n\eta_{01}\tau_{01}} \sum_{i=1}^{n} R_i (1 - T_{1i}) T_{2i}(D_i - \xi_{01}).$$

Furthermore, we have

$$\begin{aligned} \mathrm{Var}\,(RT_1(1 - T_2)(D - \xi_{10})) &= \mathrm{E}(RT_1(1 - T_2)(D - \xi_{10})^2 = \tau_{10}\eta_{10}\xi_{10}(1 - \xi_{10}) \\ \mathrm{Var}\,(R(1 - T_1)T_2(D - \xi_{01})) &= \mathrm{E}(R(1 - T_1)T_2(D - \xi_{01})^2 = \tau_{01}\eta_{01}\xi_{01}(1 - \xi_{01}) \\ \mathrm{Cov}\,[RT_1(1 - T_2)(D - \xi_{10}), R(1 - T_1)T_2(D - \xi_{01})] &= \mathrm{E}(RT_1(1 - T_1)T_2(1 - T_2)(D - \xi_{10})(D - \xi_{01}) = 0. \end{aligned}$$

Let $\boldsymbol{\xi} = (\xi_{10}, \xi_{01})^T$ and $\hat{\boldsymbol{\xi}} = (\hat{\xi}_{10}, \hat{\xi}_{01})^T$. Then $\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi})$ is asymptotically normal with mean 0 and variance-covariance matrix Σ_1, where

$$\Sigma_1 = \begin{bmatrix} \xi_{10}(1 - \xi_{10})/(\eta_{10}\tau_{10}) & 0 \\ 0 & \xi_{01}(1 - \xi_{01})/(\eta_{01}\tau_{01}) \end{bmatrix}.$$